

SOARING CLUB

FLIGHT LINE APP

Technical Specification — Version 5

React Native • Firebase Firestore • Wild Apricot Integration • iOS & Android

1. Overview

The Soaring Club Flight Line App is a real-time, multi-role mobile application coordinating ground and flight operations at the glider airfield. Version 5 refines the line-chief / no-line-chief mode toggle (owned by the tow pilot), simplifies the glider pilot's authority, introduces the pilot-entered flight time as the primary billing record with a line-chief reconciliation workflow, adds the tow pilot's tow plane selection at role login, introduces private glider shorthand display using contest number and pilot last name, and adds an end-of-day safety clearance requirement.

Roles: Glider Pilot, Tow Pilot (also controls line-chief mode), Line Chief, Bookkeeper, Admin, Ride Giver (via shared Demo Ride account). Member data is mastered in Wild Apricot and synchronized into Firebase Firestore via scheduled Cloud Functions.

2. User Roles

2.1 Glider Pilot

The glider pilot submits and manages their own tow request. The pilot has two and only two authorities over a flight ticket: Submit and Cancel. No certification or clearance step is required of the pilot.

Tow Request Form

- Student flight toggle → Solo or Dual (see Section 3 for full form logic)
- Glider selection (club or private); private gliders show N-number and contest number
- Requested tow altitude (ft AGL) and tow type
- Submit — creates the flight record; visible immediately to tow pilot and line chief
- Cancel — available any time before takeoff; archives the ticket; no other actions available to the pilot

Club Glider Flight Time Entry (Pilot-Required)

For all flights in club gliders, the pilot is required to enter their actual flight time in decimal minutes after landing. This is the primary billing record. The system enforces this progressively:

- Step 1 — In-app reminder: once a flight is marked landed (or the pilot's ticket shows airborne with no return after a threshold time), a persistent banner appears on the pilot's My Ticket screen: 'Please enter your flight time to complete your record.'
- Step 2 — Email reminder: if flight time is not entered within 2 hours of the estimated landing time, a Firebase Cloud Function sends an automated email to the pilot's address on file with a deep link back to the flight entry screen
- Step 3 — Admin flag: if flight time remains missing after 24 hours, the flight is flagged `missingFlightTime: true` and appears in the Admin's attention queue

Flight time entry is not required for private gliders. Billing for private glider flights is tow-only; the pilot's flight time is their own logbook matter.

Reconciliation with Line Chief Time

In line-chief mode, the line chief's system-calculated time (from takeoffTime to landingTime server timestamps) is a second data point. When the pilot submits their flight time, the system compares the two values:

- If the difference is within a configurable tolerance (default: ± 2.0 decimal minutes), the pilot's entry is accepted automatically and the record is marked reconciled: true
- If the difference exceeds the tolerance, both values are flagged on the line chief's screen and the Admin's attention queue as needsReconciliation: true
- Reconciliation is resolved on the field: the line chief (or Admin) reviews both values, accepts one, and marks the record reconciled with the accepted value written to billedFlightMin
- All reconciliation decisions are final on field — once reconciled: true is written, the billing value is locked and cannot be changed except by Admin with an edit reason
- In no-line-chief mode, no system-calculated time exists; the pilot's entry is the only value and is accepted automatically

2.2 Tow Pilot

The tow pilot's flight-line workflow is unaffected by whether line-chief mode is active or not. The tow pilot receives briefs and enters release altitudes regardless of mode.

Tow Plane Selection at Login

Upon selecting the Tow Pilot role at login, the tow pilot is immediately presented with a required tow plane selection screen before any other content is shown. This selection is written to their active session and is automatically applied to every flight record they certify a release altitude for.

- The tow plane selector shows all aircraft where isActive: true in the towPlanes collection
- The selected tow plane N-number is displayed persistently in the tow pilot's screen header throughout their session
- If the tow pilot changes aircraft during the day, they must log out and log back in to the Tow Pilot role to select the correct aircraft. The app prompts: 'Logging out will end your current tow plane session. Any pending release altitude entries should be completed first.'
- The tow plane selected by the tow pilot is the default pre-selection in the line chief's certification tow plane dropdown, but the line chief can override it if needed

Line-Chief Mode / No-Line-Chief Mode Toggle

The tow pilot owns the global session mode toggle. This toggle is accessible from the tow pilot's screen at any time and switches the entire app between two operating modes for all connected devices:

- Line-chief mode (default): the line chief's screen is active; the line chief logs takeoff and landing times; system-calculated flight time is available for reconciliation
- No-line-chief mode: the line chief queue screen is suppressed; takeoff and landing times are not system-logged; the pilot's self-entered flight time is the only record; no reconciliation step

- The mode is stored as a document in a globalSettings Firestore collection; all connected devices update in real time via listener
- A mode change banner appears on all connected devices when the tow pilot toggles: 'Line chief mode enabled — [tow pilot name]' or 'No line chief mode — flights self-logged by pilot'
- The tow pilot's own workflow (receive brief, enter release altitude on ground) is identical in both modes

Tow Pilot Brief Display

The tow pilot's brief card shows the following for each incoming tow:

- Club gliders: glider N-number, pilot last name, requested altitude, tow type, dual/student/passenger/demo badges
- Private gliders: contest number + pilot last name as the primary identifier (e.g. '42 / Smith'), glider N-number secondary, requested altitude, tow type, badges

End-of-Day Clearance

At the end of operations, the tow pilot is responsible for confirming that all flights have been accounted for. See Section 7 for the full end-of-day workflow.

2.3 Line Chief

Active in line-chief mode only. The line chief manages the queue, certifies flights (selects tow plane and clears for takeoff), and logs takeoff and landing times. In no-line-chief mode, the line chief screen shows a 'No line chief mode active' banner and the queue is read-only.

Queue Management

- Sees live pending queue sorted by queuePosition and an Airborne list
- Can drag-to-reorder the pending queue
- Can postpone or cancel any pending ticket

Airborne Display

All airborne flights on the line chief's screen use shorthand identifiers (see Section 6) until landing is confirmed:

- Club gliders: glider N-number + pilot last name (e.g. 'N731SG / Martinez')
- Private gliders: contest number + pilot last name (e.g. '42 / Smith')

Flight Certification

- Line chief selects the tow plane from the active roster (pre-filled from tow pilot session selection; overridable)
- Taps 'Certify — Cleared for Tow'; writes towPlaneId, towPlaneNNumber, certifiedByLineChiefId, certifiedAt; status → certified
- Log Takeoff button becomes active; line chief taps at wheels-up; status → airborne

Landing and Reconciliation

- Line chief taps Log Landing; system writes landingTime and calculates systemFlightMin (decimal minutes)
- When pilot subsequently enters their flight time, the reconciliation comparison runs automatically
- Discrepancies exceeding tolerance appear on the line chief's screen with both values for on-field resolution

2.4 Bookkeeper

Read-only access. Views billedFlightMin (the reconciled billing value) for club glider flights. Demo ride flights appear in a separate tab with ride giver name and member number.

2.5 Admin

Full database management. Owns the reconciliation tolerance setting. Reviews missingFlightTime and needsReconciliation flags in the attention queue. Can edit any flight record with a required reason.

2.6 Ride Giver / Demo Ride Account

Unchanged from v4. Demo ride account login routes directly to demo form. Ride giver is selected from the authorized list on the form. Ride giver name and member number are recorded on the flight. Waiver gate is mandatory for all demo flights.

3. Tow Request Form Logic (Glider Pilot)

3.1 Decision Tree

The form adapts through two sequential decisions. Refer to the architecture diagram for the visual decision tree.

studentFlight	isDual	Glider type	Result
true	false	Any	Student solo → soloHistory
true	true	Any	Dual instruction → instructor selector → instructionHistory
false	false	1-seat	Licensed pilot solo
false	false	2-seat	Licensed pilot solo; back seat empty
false	true	2-seat	Licensed pilot with guest passenger; passenger name required

Dual means two different things: with `studentFlight` true, Dual = CFI instruction. With `studentFlight` false, Dual = guest passenger in a two-seat club glider. These are separate code paths.

3.2 Waiver Rule

No waiver checkbox is required in the app for any regular member flight, including flights with guest passengers. The waiver gate applies only to demo ride flights.

3.3 Club Glider Flight Time Entry

After a club glider flight is marked complete or after a pilot's airborne ticket exceeds a time threshold, the pilot's My Ticket screen shows a prominent entry field: 'Your flight time (decimal minutes, e.g. 47.3)' with a Submit button. This is required for all club glider flights regardless of line-chief mode.

4. Flight Time: Pilot Entry, System Calculation, and Reconciliation

4.1 Two Time Values

For club glider flights in line-chief mode, two time values exist:

Value	Source	Purpose
<code>pilotFlightMin</code>	Pilot entry	Primary billing record; required for all club glider flights
<code>systemFlightMin</code>	Line chief timestamps	Server-calculated: $(\text{landingTime} - \text{takeoffTime}) / 60000$, one decimal
<code>billedFlightMin</code>	Reconciliation	The accepted value written after reconciliation; used for billing

4.2 Reconciliation Workflow

Step 1 — Pilot submits flight time

After landing, the pilot enters their flight time in decimal minutes. The system writes `pilotFlightMin` and triggers the reconciliation check.

Step 2 — Automatic comparison (line-chief mode only)

If `systemFlightMin` exists (line-chief mode), the Cloud Function computes: $\text{diff} = |\text{pilotFlightMin} - \text{systemFlightMin}|$

- $\text{diff} \leq \text{reconciliationToleranceMin}$ (default 2.0): auto-reconcile. `billedFlightMin` = `pilotFlightMin`; `reconciled`: true; `reconciliationMethod`: auto
- $\text{diff} > \text{tolerance}$: set `needsReconciliation`: true; flag appears on line chief screen and Admin attention queue

Step 3 — On-field resolution (if needed)

The line chief reviews both values on their device and selects the correct one. The decision is recorded with the resolver's UID and a timestamp.

- Accepts pilot's time: billedFlightMin = pilotFlightMin; reconciliationMethod: accepted_pilot
- Accepts system time: billedFlightMin = systemFlightMin; reconciliationMethod: accepted_system
- Enters a corrected value: billedFlightMin = correctedValue; reconciliationMethod: manual_correction
- reconciled: true is written; needsReconciliation: false; resolvedByUID and resolvedAt recorded

Step 4 — Final and locked

Once reconciled: true, billedFlightMin is locked. Only Admin can change it, with a required editReason and audit trail.

4.3 No-Line-Chief Mode

systemFlightMin does not exist. The pilot's entry is the only value. On submission: billedFlightMin = pilotFlightMin; reconciled: true; reconciliationMethod: no_lc_self_log. No comparison step.

4.4 Private Gliders

No flight time is required or collected for private glider flights. billedFlightMin is null. The tow charge is the only billable item. The flight record is kept for safety tracking (end-of-day clearance) but carries no billing time value.

4.5 Reminders and Escalation

Trigger	Channel	Message
Flight marked landed OR airborne > threshold	In-app banner	Persistent banner on My Ticket: 'Please enter your flight time. Required for club glider billing.'
pilotFlightMin null after 2 hrs	Email (FCF)	Automated email to pilot with deep link to flight entry screen
pilotFlightMin null after 24 hrs	Admin flag	Flight flagged missingFlightTime: true; appears in Admin attention queue

5. Tow Pilot Session: Tow Plane and Mode

5.1 Tow Plane Selection at Role Login

Immediately after the tow pilot selects the Tow Pilot role (or after login if Tow Pilot is their only role), the app presents a mandatory tow plane selection screen. No tow brief or other tow pilot content is visible until a tow plane is selected.

- Shows all towPlanes where isActive: true
- Displays make, model, and N-number for each aircraft
- Selection is confirmed with a tap; tow pilot proceeds to the tow brief screen
- Selected N-number displayed persistently in the screen header (e.g. 'Flying: N2211T')
- This selection is written to the tow pilot's session document in Firestore so the line chief's tow plane dropdown can pre-fill it

5.2 Changing Tow Planes

If the tow pilot switches to a different aircraft during the day:

- They tap their session header or a 'Change tow plane' link in the menu
- The app shows the warning: 'Logging out will end your current tow plane session. Any pending release altitude entries should be completed first.'
- On confirmation, the tow pilot is logged out of the Tow Pilot role and returned to the login screen
- They log back in, select Tow Pilot, and choose the new aircraft
- Historical flights completed during the prior session retain the original tow plane N-number

5.3 Line-Chief Mode Toggle

The tow pilot has a prominent toggle in their screen (e.g. a header switch labeled 'Line Chief On / Off'). This is a global setting stored in Firestore globalSettings.lineChiefMode (boolean).

- Default state: true (line-chief mode active)
- Toggle ON: line chief queue screen becomes active; system timestamps are recorded for takeoff and landing
- Toggle OFF: line chief queue becomes read-only with a banner; no takeoff/landing timestamps recorded; pilot self-log is the only time source
- All devices listening to globalSettings receive the mode change in real time
- A persistent mode indicator appears on every role screen when in no-line-chief mode so all users are aware
- The tow pilot's own screen layout, brief cards, and release altitude workflow are identical in both modes

The tow pilot is the natural owner of this toggle because they are always present during operations, always on the field when flying is happening, and have the operational awareness to know whether a line chief is on duty.

6. Private Glider Display and Safety Tracking

6.1 Private Glider Member Record

The privateGlider object on a member's Firestore document is updated to include the contest number:

Field	Type	Description
privateGlider.nNumber	string	FAA registration number (e.g. N5742K)
privateGlider.make	string	Manufacturer (e.g. Schempp-Hirth)
privateGlider.model	string	Model (e.g. Discus-2c)
privateGlider.contestNumber	string	2 or 3 digit competition number painted on the glider (e.g. 42, WB)

6.2 Shorthand Display Rule

For all airborne display purposes — line chief queue, tow pilot brief, and end-of-day list — private gliders use contest number + pilot last name as the primary identifier. Full details (N-number, make, model) are available on the expanded flight record but are not shown in the at-a-glance view.

Context	Club glider display	Private glider display
Tow pilot brief card	N731SG / Martinez	42 / Smith
Line chief airborne list	N731SG / Martinez	42 / Smith
End-of-day safety list	N731SG / Martinez ✓	42 / Smith ✓
Completed flight record	Full details	Full details

If a private glider member has not entered a contest number, the display falls back to N-number + last name. Admin should prompt members to add their contest number during pre-season setup.

6.3 Private Glider Safety Tracking

Private glider flight time is not billed through the club and is not required in the app. However, every private glider flight is tracked for safety: the flight record must be cleared by end of day to confirm the pilot has returned safely. This is covered in Section 7 (End-of-Day Clearance).

7. End-of-Day Safety Clearance

7.1 Purpose

At the close of each flying day, all flight records must be cleared to confirm that every pilot who flew has landed safely and is accounted for. This applies to club gliders and private gliders equally. It is a safety check, not a billing check.

7.2 Clearance Responsibility

The end-of-day clearance can be performed by either the line chief or the tow pilot. Whoever closes out the day reviews the list and marks the session complete.

7.3 End-of-Day Checklist Screen

Accessible to the line chief and tow pilot from their respective screens. Shows all flights for the current day in three categories:

Category	Meaning
Cleared ✓	Flight is complete: landed, flight time entered (if club glider), reconciled (if applicable). Pilot accounted for.
Pending action ⚠	Flight is airborne or missing pilot-entered flight time or unreconciled. Needs attention before close.
Cancelled ✕	Tow was cancelled before takeoff. No flight occurred. No further action needed.

7.4 Clearing a Flight

To move a flight from Pending to Cleared:

- If airborne (pilot has not returned): the line chief or tow pilot must physically confirm the pilot has landed safely, then mark the flight manually as 'Pilot confirmed returned safely'. This writes `manualClearanceAt` and `manualClearedByUID` to the record.
- If missing flight time (club glider, pilot has not entered time): the line chief or tow pilot may enter the flight time on the pilot's behalf with a note, or waive the time entry with an admin override and reason
- If unreconciled: resolve the time discrepancy per Section 4 before clearing

7.5 Session Close

Once all flights are in Cleared or Cancelled status, the line chief or tow pilot taps 'Close Flying Day.' This writes `dayClosedAt`, `dayClosedByUID`, and `dayClosedMode` (line_chief or tow_pilot) to a `dailySessions` document. The flying day is then locked from further queue modifications.

Private glider flights that are marked complete by the tow pilot or line chief (no flight time required) are automatically moved to Cleared status once the release altitude has been entered and the pilot is confirmed on the ground.

8. Data Model (Firestore) — Complete

8.1 globalSettings document

A single document at `globalSettings/current` shared across all devices.

Field	Type	Description
lineChiefMode	boolean	true = line chief mode active; toggled by tow pilot
modeChangedAt	timestamp	Server timestamp of last mode toggle
modeChangedByUID	string	UID of tow pilot who made the last toggle
activeTowPilotUID	string	UID of the currently logged-in tow pilot
activeTowPilotName	string	Denormalized name for display on all screens
activeTowPlaneNNumber	string	Currently selected tow plane (for line chief default)
activeTowPlaneId	string	Ref to towPlanes doc
reconciliationToleranceMin	number	Max diff before reconciliation required (default 2.0 dec min)

8.2 towPilotSessions collection

One document per tow pilot session (login to logout). Used to track tow plane changes during the day.

Field	Type	Description
sessionId	string	Auto-generated
towPilotUID	string	UID of tow pilot
towPlaneId	string	Ref to towPlanes
towPlaneNNumber	string	Denormalized tail number
sessionStart	timestamp	When tow pilot selected this plane
sessionEnd	timestamp null	When tow pilot logged out / changed plane

8.3 members collection (v5 additions)

Field (new/changed in v5)	Type	Description
privateGlider.contestNumber	string null	2 or 3 digit competition number on the aircraft

8.4 dailySessions collection

One document per flying day.

Field	Type	Description
sessionDate	string	ISO date (YYYY-MM-DD)
dayClosedAt	timestamp null	Server timestamp when flying day was closed
dayClosedByUID	string null	UID of closer (line chief or tow pilot)
dayClosedMode	string null	line_chief tow_pilot
totalFlights	number	Count of all flight records for the day

Field	Type	Description
totalCleared	number	Count of Cleared flights at close
totalCancelled	number	Count of Cancelled flights
notes	string null	Optional closer's note

8.5 flights collection (v5 additions)

Field (new/changed in v5)	Type	Description
pilotFlightMin	number null	Pilot-entered flight time in decimal minutes
pilotFlightMinAt	timestamp null	When pilot submitted their time
systemFlightMin	number null	Server-calc (LC mode only): (landingTime-takeoffTime)/60000
billedFlightMin	number null	Final accepted value used for billing; null for private gliders
reconciled	boolean	True once billing value is final
reconciliationMethod	string null	auto accepted_pilot accepted_system>manual_correction no_lc_self_log
needsReconciliation	boolean	True when diff > tolerance and not yet resolved
resolvedByUID	string null	UID of resolver (LC or Admin)
resolvedAt	timestamp null	When reconciliation was resolved
missingFlightTime	boolean	True after 24hrs with no pilot entry for club glider
reminderEmailSentAt	timestamp null	When 2hr reminder email was sent
manualClearanceAt	timestamp null	Set when LC/TP manually confirms pilot returned safely
manualClearedByUID	string null	UID of the manual clearer
towPilotSessionId	string	Ref to towPilotSessions doc (links tow plane to flight)
displayShorthand	string	Pre-computed: e.g. '42 / Smith' or 'N731SG / Martinez'; used in all list views

9. Complete Workflow (v5)

9.1 Start of Day

- Tow pilot logs in, selects Tow Pilot role, selects tow plane — session begins
- Tow pilot confirms or toggles line-chief mode (default: on if a line chief is present)
- Line chief logs in if present; sees mode confirmed active
- All devices show active tow plane in header

9.2 Per-Tow Workflow (Line-Chief Mode)

Step	Actor	Action
1	Glider pilot	Submits tow request (student/dual/passenger/altitude/type)
2	Tow pilot	Brief appears in real time; reviews (no action required)
3	Line chief	Reviews queue; selects tow plane; taps Certify — Cleared for Tow
4	Line chief	Taps Log Takeoff at wheels-up; status → airborne
5	Tow pilot	Tows glider; no app action required during flight
6	Tow pilot	Returns to ground; enters release altitude on completed tow card
7	Line chief	Taps Log Landing when glider touches down; systemFlightMin calculated
8	Glider pilot	Enters pilotFlightMin on My Ticket (club glider required)
9	System	Reconciliation comparison runs; auto-reconciles or flags for LC review
10	Line chief (if needed)	Resolves discrepancy on-field; billedFlightMin locked

9.3 Per-Tow Workflow (No-Line-Chief Mode)

Step	Actor	Action
1	Glider pilot	Submits tow request
2	Tow pilot	Brief appears; reviews; tows; enters release altitude on ground
3	Glider pilot	Enters pilotFlightMin after landing (club glider required)
4	System	billedFlightMin = pilotFlightMin; reconciled: true; no comparison

9.4 End of Day

- Line chief or tow pilot opens End-of-Day Checklist
- Resolves any Pending Action items (missing time, unreconciled, still-airborne)
- Confirms all private glider pilots have returned safely
- Taps Close Flying Day; dailySessions document written

10. Screen Map (Complete)

Screen	Role(s)	Purpose
Login	All	Email/password; demo account routes direct to demo form
Role Select	Multi-role	Choose session role

Screen	Role(s)	Purpose
Tow Plane Select (gate)	Tow Pilot	Mandatory before any tow pilot content; shows active aircraft
Tow Brief (live)	Tow Pilot	Incoming briefs; shorthand IDs; release alt entry on ground
Mode Toggle (header)	Tow Pilot	Line Chief On/Off; updates globalSettings in real time
End-of-Day Checklist	Tow Pilot, LC	All flights: Cleared / Pending / Cancelled; Close Flying Day
Tow Request Form	Glider Pilot	Student/dual branch; glider; altitude; tow type; Submit/Cancel only
My Ticket	Glider Pilot	Status; queue position; flight time entry (club glider); cancel
Demo Ride Form	Demo Acct	Ride giver select; passenger; waiver gate; two-seat glider
Flight Queue (LC mode)	Line Chief	Pending (reorder, certify+plane, clear) + Airborne (shorthand IDs)
Reconciliation Panel	Line Chief	Shows both times; select accepted value; locks billing
Flight Queue (no-LC mode)	Line Chief	Read-only with mode banner
Daily Log — Flights	Bookkeeper	billedFlightMin per flight; instructor info; all fields
Daily Log — Demo Rides	Bookkeeper	Ride giver name+number; passenger; billing notes
Student History	BK + Admin	Dual/solo decimal hours; instructors; tow types
Admin: Attention Queue	Admin	missingFlightTime + needsReconciliation flags
Admin: Members	Admin	All flags; CFI#; contest numbers; demo account; WA sync
Admin: Aircraft	Admin	Gliders (isTwoSeat) and tow planes (isActive)
Admin: Flight Edit	Admin	Any field; reason required; audit trail
Member Profile	All	Name; member#; private glider N-number + contest number

11. Technology Stack

Layer	Technology
Mobile framework	React Native (Expo) + TypeScript
Navigation	React Navigation v6; role-based routing; tow plane gate screen
Database	Firebase Firestore; real-time listeners; globalSettings for mode sync
Auth	Firebase Auth; email/password; demo account detection

Layer	Technology
Cloud Functions	WA sync; onFlightComplete (history, reconciliation check); reminder email trigger; day-close calc
Email reminders	Firebase Cloud Functions + SendGrid (or Firebase Email Extension) for 2hr missing-time reminder
WA integration	Wild Apricot REST API v2.1; OAuth2; Secret Manager
Drag-to-reorder	react-native-draggable-flatlist; atomic batch write
Forms	React Hook Form; decimal minute validation; waiver gate
Offline	Firestore enablePersistence(); offline banner
Export	csv-stringify; react-native-pdf-lib
Push notifications	FCM — tow pilot alert; mode change notification
CI/CD	Expo EAS Build; OTA updates

12. Development Phases

Phase 1 — Core MVP (7–9 weeks)

- Firebase; full schema; security rules; globalSettings listener
- Tow pilot: tow plane gate screen at login; mode toggle; passive brief; release alt on ground
- Glider pilot: tow form (student/dual branch); Submit and Cancel only; club glider flight time entry with in-app reminder
- Line chief: queue with drag-to-reorder; certification (tow plane select + Certify); Log Takeoff / Landing; reconciliation panel
- Reconciliation Cloud Function; auto-reconcile; needsReconciliation flag
- Demo ride form via shared account; ride giver selector; waiver gate
- Shorthand display for all airborne cards (contest# / last name)
- End-of-day checklist and Close Flying Day

Phase 2 — Instruction + Reminders (3–4 weeks)

- onFlightComplete Cloud Function: instructionHistory / soloHistory subcollections
- Student History screen
- 2hr reminder email via Cloud Function + SendGrid
- missingFlightTime flag and Admin attention queue

Phase 3 — Wild Apricot (3–4 weeks)

- Scheduled WA sync; app-only flags never overwritten; contest number not in WA so protected from sync
- Admin: all flag management; WA sync log; demo account setup
- Membership gate at login

Phase 4 — Polish & Export (2–3 weeks)

- CSV/PDF export; FCM push; offline hardening; admin flight edit

Phase 5 — Enhancements

- WA billing write-back; instructor endorsement PDF; solo auth workflow; METAR; winch launch
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13. Notes

The reconciliation tolerance of 2.0 decimal minutes is a suggested default. The operations manager should review this value with the chief flight instructor and bookkeeper before go-live. A tighter tolerance (e.g. 1.0) creates more on-field reconciliation events; a looser one (e.g. 5.0) risks billing inaccuracy. The tolerance is stored in `globalSettings` and can be adjusted by Admin without a code change.

The tow pilot logout-and-back-in workflow for changing aircraft is deliberate. It creates a clean session boundary in `towPilotSessions`, ensuring every historical tow record has an unambiguous link to the correct tow plane via `towPilotSessionId`. An in-app aircraft swap without logout would complicate the session/plane linkage and make audit trails harder.

Contest numbers are not managed by Wild Apricot. They are entered by the pilot via their Member Profile screen or by Admin. The WA sync must never overwrite the `privateGlider` object. This is enforced in the sync Cloud Function by checking whether the incoming WA record contains a `privateGlider` field (it will not) and skipping that field entirely.

The end-of-day manual clearance for private gliders ('Pilot confirmed returned safely') is intentionally a human-confirmed action, not an automatic one. It ensures a person on the field has physically verified or communicated with the pilot — the app supports but does not replace that human check.

This document should be reviewed with the club's tow pilots, line chiefs, operations manager, and bookkeeper before development begins.

Application Testing

- **iOS testing:** Use **TestFlight**
 - **Android testing:** Use **Google Play Internal Testing**
 - No public release required
 - Only people you invite can install the test builds
 - Works perfectly with apps generated by Claude, Copilot, or any other AI coding assistant
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iOS Testing (TestFlight)

TestFlight is Apple's official beta-testing platform. It lets you install and test apps **without publishing them**.

How it works

1. You (or your developer) upload the app build to App Store Connect.
2. Add testers using their **email addresses** or share a private invite link.
3. Testers install the **TestFlight app** from the App Store.
4. They can then install your TSA app directly from TestFlight.

Who can test

- Up to **100 internal testers** (using your Apple Developer account)
- Up to **10,000 external testers** (via email or link)

Why this is ideal

- No public visibility
 - No App Store review required for internal testers
 - Easy to push new builds as you iterate with Claude
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Android Testing (Google Play Internal Testing)

Google provides a similar system.

How it works

1. Upload the Android APK/AAB to Google Play Console.
2. Create an **Internal Test Track**.
3. Add testers by email or share a private link.
4. Testers install the app through Google Play, but it's **not public**.

Who can test

- Up to 100 internal testers
- Closed testing tracks allow more if needed

Why this is ideal

- Fast deployment
 - No public listing
 - Easy updates
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How you test while building with Claude

Claude can generate:

- iOS Swift/SwiftUI code
- Android Kotlin/Jetpack Compose code
- React Native or Flutter code (one codebase for both platforms)

You then:

1. Build the app locally or in a cloud CI service
2. Upload builds to TestFlight and Google Play Internal Testing
3. Install on your own devices
4. Iterate quickly

This is exactly how professional teams test apps before release.

How to keep testing limited to TSA

You control access by:

iOS

- Only invite TSA members via email
- Or share a private TestFlight link only with TSA
- App is not visible in the App Store

Android

- Only invite TSA members via email
- Or share a private Google Play testing link
- App is not visible publicly

Inside the app

You can add a TSA login screen so even if someone gets the link, they can't use the app.

Recommended TSA workflow

1. Build the app with Claude (or any AI).
2. Test internally using TestFlight + Google Internal Testing.
3. Iterate until stable.
4. When ready for members:
 - Publish privately via Apple Business Manager + Managed Google Play
 - Or use unlisted distribution + TSA login

This gives you **full control** and avoids public exposure.

If you want, I can map out the exact step-by-step setup for:

- Apple Developer account
- TestFlight configuration
- Google Play Console setup
- How to export builds from Xcode or Android Studio
- How to structure the TSA login system

Which part do you want to tackle next?